26F Stat TA Session W14

0. Preparation for TA Session

- Download the sample code and and attached dataset `mlb_players_18` and `possum` (click here)
- Import the two files as `mlb` and `possum` , respectively, using the code below
  - `mlb` contains batter statistics for 2018 MLB season
  - `possum` records data on possums (a rat-like animal) in Australia, including site, population, sex, age, head length, and tail length

```
library(dplyr)

# we want to analyze batter performance
mlb <- read.csv("data/mlb_players_18.csv") %>%
    filter(position != "P") %>% # Pitcher
    filter(position != "DH") # Designated Hitter

possum <- read.csv("possum.csv")
```

1. Analysis of Variance (ANOVA) `aov()`

- Recall we have learned two types of tests so far:
  - Chi-square test examines if categorical variables are associated
  - t-test compares the mean for two numerical variables
- ANOVA is like an extension of t-test, which is used for comparison of the means for multiple numerical variables
  - For example, we may wonder: if the average performance (`OBP`) differs for batters in different `positions`

```
# syntax
# perform ANOVA
aov_result <- aov(data$ValueVar ~ data$GroupVar)
# show result
summary(aov_result)

# example
aov_result <- aov(mlb$OBP ~ mlb$position)
summary(aov_result)
```

```
# output
              Df Sum Sq  Mean Sq F value  Pr(>F)
mlb$position   7 0.1074 0.015346   3.129 0.00297 **
Residuals    614 3.0118 0.004905
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

2. Linear Regression `lm()`

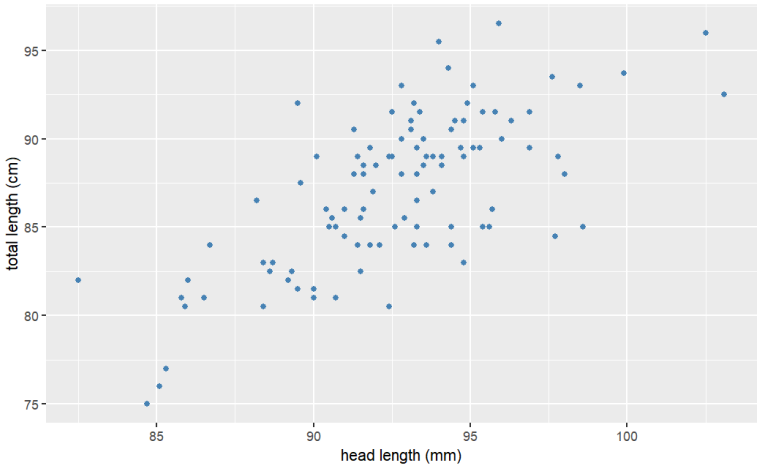


Figure: Head length and total length of possums in Australia

- In most economics subfields, even in machine learning, linear regression is a very powerful tool to fit data with our theoretical-grounded models (e.g., Solow model in Macroeconomics)
- The idea is rather simple, as you may have learned in high school. We want to find a straight line that minimizes the sum of square errors (i.e., distance to sample points)

```
# syntax
lm(myData$var1 ~ myData$var2)
lm(data = myData, var1 ~ var2 )

# example
lm_model <- lm(possum$total_l ~ possum$head_l)
lm_model <- lm(data = possum, total_l ~ head_l)
print(lm_model)

# output
Call:
lm(formula = total_l ~ head_l, data = possum)

Coefficients:
```

```
(Intercept)       head_l
     9.8882       0.8337
```

- The regression outcome finds the linear regression:

$$\widehat{\text{total length}} = 9.8882 + 0.8337 \cdot \text{head length}$$

Obtain Regression Results `summary()`

- Print `lm_model` only shows the estimated coefficients, but we care more than these!
- `lm()` fits the linear model
- `summary()` analyzes and reports that fit by showing p-values, R^2, etc. These are not computed in `lm()`

```
summary(lm_model)

# output
Call:
lm(formula = total_l ~ head_l, data = possum)

Residuals:
    Min      1Q  Median      3Q     Max
-7.0881 -2.2935  0.2888  2.0801  7.4983

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.88823    8.00016   1.236    0.219
head_l       0.83367    0.08633   9.657 4.68e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.131 on 102 degrees of freedom
Multiple R-squared:  0.4776,    Adjusted R-squared:  0.4725
F-statistic: 93.26 on 1 and 102 DF,  p-value: 4.681e-16
```

Adding Fitted Line to the plot `geom_smooth()`

- We use scatter plot to explore the relationship between head and total lengths
- Adding the module `geom_smooth()` puts the fitted line on our initial scatter plot

```
# syntax
Your_Plot +
    geom_smooth(method = "lm", se = TRUE)
# method: the method to smooth the data
```

```r
# se: whether to display confidence interval (default = TRUE))

# example
ggplot(possum, aes(x = head_l, y = total_l)) +
  geom_point(color = "steelblue") +
  geom_smooth(method = "lm",
              se = TRUE) +
  labs(x = "head length (mm)",
       y = "total length (cm)")
```

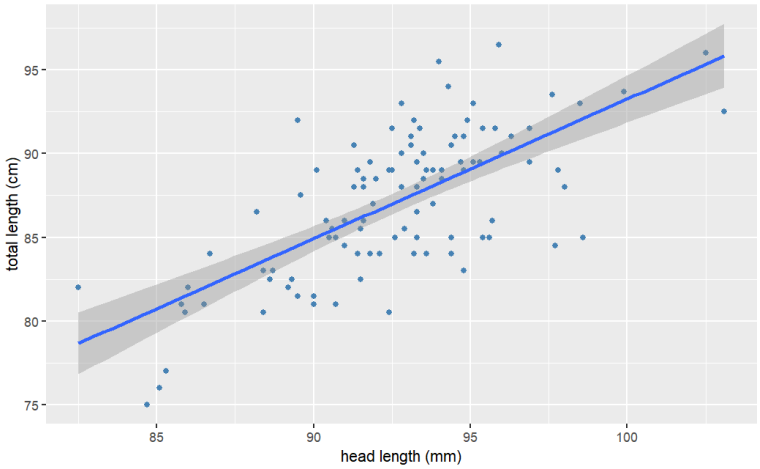


Figure: head length and total length of possums in Australia, with fitted line

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Exercises

W14 Exercises